Grade: Check Plus

Assignment 2 02/24/2026

Renan Camarotti

A stock has a current selling price of $20. The distribution of the price of the stock one year from now is believed to be as follows:

Table 1

PRICE (x)	f(x)	x f(x)	$(x-\mu)^2$ f(x)
20	0.05	1.00	1.46
21	0.06	1.26	1.17
22	0.07	1.54	0.81
23	0.08	1.84	0.46
24	0.09	2.16	0.18
25	0.12	3.00	0.02
26	0.15	3.90	0.05
27	0.13	3.51	0.33
28	0.1	2.80	0.67
29	0.1	2.90	1.29
30	0.05	1.50	1.05
∑	1	25.41	7.50

An investor holds a one year option (European) on this stock. This enable him to purchase, one year from now, one share of the stock at $25 per share. The value of this option will depend upon the price of the stock one year from now. If the price rises to $28, the value of the option (ignoring interest and commissions) is $3. If the price rises to only $23, the option is worthless.

1. Compute the mean and standard deviation of the stock price.

✓ Q1

Mean (μ) = ∑ x f(x) = **$25.41** (as calculated in Table 1 above)

Standard deviation (σ) = $[\sum (x- \mu)^2 f(x)]^{1/2} = [7.50]^{1/2}$ = **$2.74** (Variance calculated in Table 1 above)

2. a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?

✓ Q2a

P(μ-σ ≤ D ≤ μ-σ) = P(25.41 – 2.74 ≤ D ≤ 25.41 + 2.74) =

P(22.67 ≤ D ≤ 28.15) = P(23 ≤ D ≤ 28) = f(23) + f(24) + f(25) + f(26) + f(27) + f(28) =

0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.1= **0.67 or 67%**

b. What is the probability that the stock price will fall within 2 standard deviation of the mean one year from now? ✓ Q2b

$P(\mu\text{-}2\sigma \le D \le \mu\text{-}2\sigma) = P(25.41 - 2x2.74 \le D \le 25.41 + 2x2.74) =$

$P(19.93 \le D \le 30.89) = P(20 \le D \le 30) =$ **1.00 or 100%**

3. Derive the distribution of the option value, ignoring interest and commissions.

Table 2

Option value (y)	f(y)	y f(y)	$(y\text{-}\mu)^2$ f(y)
0.00	0.05	0.00	0.09
0.00	0.06	0.00	0.11
0.00	0.07	0.00	0.13
0.00	0.08	0.00 ✓ Q3	0.15
0.00	0.09	0.00	0.17
0.00	0.12	0.00	0.22
1.00	0.15	0.15	0.02
2.00	0.13	0.26	0.05
3.00	0.10	0.30	0.27
4.00	0.10	0.40	0.70
5.00	0.05	0.25	0.66
∑	1	1.36	2.57

4. What is the probability that the option will have positive value? ✓ Q4

$P(Op \ge 0) = f(1.00) + f(2.00) + f(3.00) + f(4.00) + f(5.00) =$

$0.15 + 0.13 + 0.10 + 0.10 + 0.05 =$ **0.53 or 53%**

5. Compute the mean and standard deviation of the value of the option.

Mean (μ) = $\sum x\, f(x)$ = **$1.36** (as calculated in ✓ Q5 Table 2 above)

Standard deviation (σ) = $[\sum (x\text{-}\mu)^2 f(x)]^{1/2} = [2.57]^{1/2}$ = **$1.60** (Variance calculated on Table 2 above)